Java Programming

Unit-wise Important Questions — No Repetition

Prepared by comparing: Question Bank PDF, 60 Important Questions PDF, and the supplied Parul University Java syllabus image.

How to use this list: Questions with overlapping topics have been merged; distinct programming tasks are retained. This is a revision guide based on the supplied material, not a guarantee of exact exam questions.

Unit 1: Overview of OOP and Java

Syllabus weightage: 10% | Teaching hours: 4

1. Define OOP and explain its main features. Explain the four pillars of OOP with examples.

2. Differentiate between procedural programming and object-oriented programming.

3. Explain the history and important features of Java.

4. Explain JVM architecture with a neat diagram.

5. Differentiate between JDK, JRE and JVM.

6. Explain the structure of a basic Java program and the compilation and execution process.

7. Explain PATH and CLASSPATH in Java. What is an IDE? Give examples.

Unit 2: Java Language Basics

Syllabus weightage: 15% | Teaching hours: 7

1. Explain variables and their types, primitive and reference data types in Java.

2. Explain type casting. Differentiate between implicit (widening) and explicit (narrowing) casting with examples.

3. Explain wrapper classes, autoboxing and unboxing with examples.

4. Explain different types of operators and operator precedence in Java.

5. Explain if, if-else, nested-if and switch statements with examples.

6. Differentiate between for, while and do-while loops.

7. Explain the enhanced for-each loop with an example.

8. Differentiate between break and continue statements with examples.

Unit 3: Arrays and String Arrays

Syllabus weightage: 12% | Teaching hours: 6

1. Explain one-dimensional and multidimensional arrays with examples.

2. Write a Java program to find the second-largest element in an array without using built-in sorting methods.

3. Write a Java program to find the largest element in a two-dimensional array.

4. Explain common array operations: traversal, searching, sorting, insertion, deletion and updating.

5. Explain the String class and its commonly used methods.

6. Differentiate between String, StringBuffer and StringBuilder.

7. Explain how strings are compared using equals(), == and compareTo().

Unit 4: Object-Oriented Programming (OOP)

Syllabus weightage: 30% | Teaching hours: 12

1. Explain classes, objects and constructors. Describe no-argument and parameterized constructors.

2. What is constructor overloading? Explain constructor chaining using this and super with examples.

3. Explain access specifiers in Java.

4. Explain encapsulation with a suitable Java program.

5. Explain inheritance and its types with examples. Why does Java not support multiple inheritance through classes?

6. Explain the this, super and final keywords with examples.

7. Explain polymorphism, method overloading and method overriding. Differentiate between overloading and overriding.

8. Write a Java program demonstrating inheritance, constructor chaining using super and method overriding.

9. Explain abstraction and abstract classes. Differentiate between abstract classes and interfaces.

10. What is an interface? Explain functional interfaces with examples.

11. Explain the Object class and its important methods.

Unit 5: Exception Handling

Syllabus weightage: 15% | Teaching hours: 6

1. Define exceptions. Differentiate between errors and exceptions.

2. Differentiate between checked and unchecked exceptions with examples.

3. Explain try-catch-finally and multiple catch blocks with a suitable example.

4. Differentiate between throw and throws keywords with examples.

5. What is a custom exception? Write a Java program for InvalidAgeException when age is below 18.

6. Explain exception-handling best practices.

Unit 6: Collection Framework in Java

Syllabus weightage: 10% | Teaching hours: 5

1. Explain the Java Collections Framework and its main interfaces.

2. Explain the List interface. Differentiate between ArrayList and LinkedList and demonstrate ArrayList methods.

3. Explain the Set interface. Differentiate between HashSet and TreeSet.

4. Explain the Map interface. Differentiate between HashMap and TreeMap. Write a program to store and retrieve student records using HashMap.

5. Explain iterators and generics. Write a generic class to swap two elements and explain the benefits of generics.

6. Write a Java program to remove duplicate elements from a list and display unique elements in sorted order.

Unit 7: Advanced Foundations and Java Frameworks

Syllabus weightage: 8% | Teaching hours: 5

1. Differentiate between a process and a thread.

2. Explain multithreading, the two ways to create threads, and the thread life cycle.

3. Explain thread synchronization. Write a program using two Runnable threads and a shared counter.

4. Explain inter-thread communication using wait(), notify() and notifyAll().

5. Explain file handling in Java and the File class. Differentiate between byte streams and character streams.

6. Write a Java program to write data to and read data from a file using FileOutputStream and FileInputStream. Explain buffered streams.

7. Explain serialization and deserialization. Describe Serializable, ObjectOutputStream and ObjectInputStream.

8. Explain lambda expressions and functional interfaces with examples.

9. Explain the Stream API and its filter(), map(), sorted(), collect() and forEach() operations. Write a program to filter even numbers and calculate their squares.

10. Explain Optional in Java with an example. Use Stream API and Optional to process student marks and safely handle a missing student.

Total: 55 consolidated questions across 7 units.